MILITARY SPECIFICATION

PROJECTILE, 155MM, HE: XM864
LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by the US Army AMCCOM, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of projectile designated as Projectile, 155MM, HE: XM864.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-P-48099	- Propellant, M10 (For use in Special Applications)

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-286 - Propellants, Solid; Sampling, Examination and Testing
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-453 - Inspection, Radiographic
MIL-STD-1235 - Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials; Test Procedures

DRAWINGS (See 6.6)

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

PRODUCT AND PACKING DRAWINGS

9381131 - Projectile, 155MM, HE: XM864
8837839 - Pallet for Projectile, 155MM, HE: and GB, Hollow Base Type

INSPECTION EQUIPMENT DRAWINGS

8841732 - Test Fixture
9202990 - Operating Instructions

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

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(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Projectile. The projectile assembly and all components thereof shall comply with all requirements specified on Drawing 9381131, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Strength of heat seal. The charge bag shall withstand application of a static load, at the seam, of 2.5 pounds per linear inch, minimum without peel-back of the heat seal and without tensile failure of the area adjacent to the heat seal.

3.4 Ballistic functioning.

3.4.1 Projectile. The projectile shall function satisfactorily.

3.4.1.1 Premature burst. The projectile shall not prematurely burst. A premature burst is considered to have occurred if the projectile functions in the weapon or if the time of flight is less than 1.9 seconds.

3.4.1.2 Functioning at all zones, excluding safety phase. Satisfactory functioning of the projectile when fired at any zone, except the safety phase, is defined as successful delivery of the cargo to the target area with all projectile hardware remaining intact until initiation of the time fuze at a pre-determined time after firing, and shall also include the successful expulsion of the entire cargo of grenades.

3.4.1.3 Functioning at the safety phase. Satisfactory functioning of the projectile when fired at the safety phase is defined as successful delivery of the cargo to the target area with all projectile hardware remaining intact until initiation by the time fuze.

3.4.1.4 Range spread. The total variation in range of airburst, when the time fuze properly functions, shall not exceed 800 meters.

3.4.2 Grenade. The grenade shall function satisfactorily.

3.4.2.1 Unsatisfactory functioning. Grenades failing to function satisfactorily are defined as those grenades which upon ground impact do not function high order, except those grenades evaluated to have been damaged by air collision/detonation.

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3.4.2.2 Air detonation. A grenade air detonation is defined as functioning of the grenade in the air due to any cause during the time interval after expulsion from the projectile and prior to impact on the prepared field.

3.4.3 Safety. There shall be no premature burst or evidence thereof, and there shall be no separation of metal parts either in the bore of the weapon or in flight (prior to initiation of the time fuze) in any of the ballistic functioning tests. This requirement shall also apply to those rounds to be fired in the safety phase.

3.5 Radiographic examination of projectiles. Prior to forwarding the projectiles to the proving ground for ballistic testing, the projectiles shall be subjected to radiographic examination. The projectiles shall be examined for the following:

a. Gap between adapter and base. (see note)
b. Gap between adapter and grenade. (see note)
c. Excessive gap at take-up pad area between grenade and forward plate. (see note).
d. Gap between grenades. (see note)
e. Safety pin or safety clip shall not be present.
f. Any missing or improperly positioned component or assembly normally discernible on the radiograph.
g. Separated ignitor components.

Note: Gaps in excess of 0.015 inches shall be considered rejects for this inspection.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Records of results of contractor-conducted examinations and tests. The contractor shall maintain records of the results of all examinations or tests required by this specification. (see 6.5)

3.8 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. In addition, plastic component parts shall be free from splits, cold flow, shrinkage, flash, gouges, cold shuts, inclusion, porosity, or any similar characteristic. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material and protective coating missing per paragraph 4.4.2.14 are permitted. All required marking and stamping shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Lid Container, Expulsion Charge	9362552	See Note
Cup Container, Expulsion Charge	9362553	See Note
Projectile	9381131	12
Steel Shim	9215397	25
Expulsion Charge Assembly	9272019	50
Container Assembly Expulsion Charge	9362555	50
Expulsion Charge Assembly	9362556	25
Obturator	10542907	5

Note: 5 samples from each mold cavity.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Projectile and Components				NEXT HIGHER ASSEMBLY: See Below
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile (Dwg 9381131)				
	Examination for Defects	12	-	3.2	4.4.2.14
	Ballistic functioning	12	-	3.4	4.4.3.1
	Expulsion Charge Assembly (Dwg 9272019)				
	Examination for Defects	50		3.2	4.4.2.4
	Steel Shim (Dwg 9215397)				
	Examination for Defects 4.4.2.7	25		3.2	4.4.2.7
	Lid Container, Expulsion Charge (Dwg 9362552)				
	Examination for Defects	(a)		3.2	4.4.2.17
	Cup Container, Expulsion Charge (Dwg 9362553)				
	Examination for Defects	(a)		3.2	4.4.2.18
	Container Assembly Expulsion Charge (Dwg 9362555)				
	Examination for Defects	50		3.2	4.4.2.20
	Explusion Charge Assembly (Dwg 9362556)				
	Examination for Defects	25		3.2	4.4.2.21

NOTES:
(a) Five (5) samples from each mold cavity.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70507 (AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile and Components		SHEET 2 OF 2		See Below
	Obturator (Dwg 10542907) Examination for Defects	(a)		3.2	4.4.2.16
	Forward Plate (Dwg 9381111) Examination for Defects	10		3.2	4.4.2.24

NOTES:

(a) Five (5) samples from each mold cavity.

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4.3.3 Rejection. See MIL-A-48078. In addition, the first article sample shall be rejected if the ballistic test results reveal any one of the following:

a. Any premature burst occurs.
b. Any separation of metal parts occurs prior to initiation of the time fuze.
c. Any detonation of the projectile cargo occurs prior to initiation of the time fuze.
d. The total number of grenades failing to function satisfactorily during the service pressure phase equals or exceeds twenty-nine (29).
e. Any projectile fails to function satisfactorily.
f. The variation in range to point of air burst exceeds 800 meters.

Note 1: A round-by-round accounting shall be made for lost grenades and the replacement provisions of 4.4.3.1.1 utilized when applicable.

Note 2: Grenade air detonations shall be counted and recorded for informational purposes. However, if the total number of air detonations from the entire eight rounds tested in service pressure phase exceeds thirteen (13), the data shall be referred to the engineering agency (See 6.4) for decision on final acceptance of the first article.

Note 3: Grenades evaluated to have malfunctioned due to damage resulting from air collision/detonation shall not be included in the total number of grenades used to accept or reject the lot. Grenades so malfunctioning shall be counted and recorded on the evaluation sheet for informational purposes. However, if the total number of such grenades from the entire eight (8) rounds tested in the service pressure phases exceeds sixteen (16), the data shall be referred to the engineering agency (See 6.4) for decision on final acceptance of the first article

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Grenades (M42) from one finally accepted lot interfix number from one manufacturer (See Note 1).

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b. Grenades (M46) from one finally accepted lot interfix number from one manufacturer (See Note 1).
c. Projectile metal parts (body/ogive assembly) from one lot interfix number from one manufacturer.
d. Base burner assemblies from one lot, from one interfix number and from one manufacturer.
e. Adapters from one lot interfix number from one manufacturer.
f. Forward plates from one lot interfix number from one manufacturer.
g. Sleeves from one lot interfix number from one manufacturer.
h. Spacers from one lot interfix number from one manufacturer.
i. Splines from one lot interfix number from one manufacturer.
j. Charge assembly propellant from one manufacturer and no more than two serial lot numbers.
k. Expulsion charge assemblies from one lot interfix number from one manufacturer.
l. Loaded projectiles of one weight zone (See note 2).
m. Lids, explusion charge container from one lot interfix number from one manufacturer.
n. Cup, expulsion charge container from one lot interfix number from one manufacturer.

Note 1: Only those grenade lots which have satisfactorily passed the grenade functioning test shall be assembled into the projectile.

Note 2: Projectiles falling out of the inspection lot criteria due to the zone weight restriction may be accumulated into a hybrid lot provided they originally met all other lotting restrictions during manufacture and the original lot passed all acceptance tests. Further ballistic testing of this hybrid lot is not required.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be utilized if approved by the Procuring Activity.

b. Automatic gage. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with the contract.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272021
4.4.2.1	Bag, Charge				NEXT HIGHER ASSEMBLY 9272020
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of bag		0.40%	3.2	Gage
102	Distance to end heat seal, max.		0.40%	3.2	Gage
103	Distance to side heat seal		0.40%	3.2	Gage
104	Minimum width of both heat seals over entire length		0.40%	3.2	Gage
105	Heat seal incomplete (2 places)		0.40%	3.2	Visual
106	Strength of heat seal	(a)		3.3	4.5.3
107	Location of eyelet		0.40%	3.2	Gage
108	Location of end stitching		0.40%	3.2	Gage
109	Distance to side stitching		0.40%	3.2	Gage
Minor					
201	Stitching incomplete (2 places)		0.65%	3.2	Visual
202	Fold width eyelet end		0.65%	3.2	Gage
203	Fold misalignment max.		0.65%	3.2	Gage
204	Thickness of assembled eyelet		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample 20 per lot, reject on one (1) or more defectives.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272022
4.4.2.2	Cover, Expulsion Charge				NEXT HIGHER ASSEMBLY 9272020
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of rim		0.40%	3.2	Gage
102	Thickness of rim, minimum		0.40%	3.2	Gage
103	Total height		0.40%	3.2	Gage
104	Angle between rim and angled portion		0.40%	3.2	Gage
105	Thickness of angled portion, minimum		0.40%	3.2	Gage
106	Height of rim		0.40%	3.2	Gage
107	Location of tabs		0.40%	3.2	Gage
108	Distance between tab and angled portion (4 places)		0.40%	3.2	Gage
109	Radius missing		0.40%	3.2	Visual
110	Crack or puncture		0.40%	3.2	Visual
111	Burr		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272020
4.4.2.3	Charge Loading Assembly				NEXT HIGHER ASSEMBLY 9272019
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pull wire or bag loose with respect to cover		0.40%	3.2	Visual/Manual
102	Cover damaged		0.40%	3.2	Visual
103	Clinch of stud improper		0.40%	3.2	Visual
104	Bag damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272019
4.4.2.4	Expulsion Charge Assembly				NEXT HIGHER ASSEMBLY 9381131
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leak test	(a)		3.2	4.5.2
102	Weight of propellant charge incorrect		0.40%	3.2	Gage
103	Bag seal incomplete (3 places)		0.40%	3.2	Visual
104	Moisture and total volatiles content of propellant	(a)		3.2	4.5.1
105	Bag damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample in accordance with applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215380
4.4.2.5	Adapter				NEXT HIGHER ASSEMBLY 9381131
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical I	Verification of heat treatment of adapter		100%	3.2	4.5.4
Major	None defined				
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Projectile Metal Parts Assembly Prior to Loading	1 OF 1	9381117
			NEXT HIGHER ASSEMBLY 9381131

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Threads damaged at base end		0.40%	3.2	Visual
102	Threads damaged at nose end		0.40%	3.2	Visual
103	Internal lubricant applied improperly		0.40%	3.2	Visual
Minor					
201	Foreign matter on interior surface of body		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Shim, Steel				9215397
					NEXT HIGHER ASSEMBLY 9381131
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness		0.40%	3.2	4.5.5
102	Diameter, min.		0.40%	3.2	Gage
103	Thickness, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Assembly, Forward Plate and O-Ring		9381111
			NEXT HIGHER ASSEMBLY 9381131

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	O-ring not greased		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-P-70507 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Grenade Assembly, Prior to Loading Into Projectile		9215340 or 9215370 as applicable
			NEXT HIGHER ASSEMBLY 9381131

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety pin missing		100%	3.2	Visual
2	Slider lock cracked, incomplete, missing or improperly positioned		100%	3.2	Visual
Major					
101	Safety clip not removed		0.40%	3.2	Visual
102	Ribbon unwound		0.40%	3.2	Visual
103	Tape not completely inserted in slots of stiffener		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.10	Projectile, After Removal of Safety Pins From Grenade Assembly, After each layer is inserted into the projectile		1 1		9381131
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Slide assembly protruding from grenade fuze housing indicating armed or partially armed (all layers)		100%	3.2	Visual
2	Grenade type incorrect as evidenced by protective coating or lack of yellow marking (layers (7-9) only)		100%	3.2	Visual
3	Slider lock cracked, incomplete, missing or improperly positioned		100%	3.2	Visual
4	Grenade missing or improperly positioned (all layers)		100%	3.2	Visual
5	Any sleeve, spacer or spline missing, improperly positioned or incorrect length (all layers)		100%	3.2	Visual
Major					
101	Grenade type incorrect (1-6 layers only)		0.40%	3.2	Visual
102	Safety pin not removed from each grenade (all layers)		0.40%	3.2	Visual
103	Stiffener or tape improperly positioned (all layers)		0.40%	3.2	Visual
Minor	None defined				
NOTES:					

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH 4.4.2.11	TITLE Projectile after Assembly of Adapter and Immediately Prior to Assembly of Base Burner Assy and Seal Ring		1 SHEET 1 OF		DRAWING NUMBER 9381131
					NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Adapter missing or improperly assembled		100%	3.2	Visual
2	Any sleeve, spacer or spline missing, improperly positioned or incorrect length		100%	3.2	Visual
Major					
101	Depth between projectile face and adapters or shims as applicable at specified load		100%	3.2	4.5.11
102	Seal ring missing or wrong type		0.40%	3.2	Visual
103	Seal ring not greased		0.40%	3.2	Visual
104	Sealant missing on threads of base burner assy.		0.40%	3.2	Visual
105	Threads damaged on base burner assy. of projectile body		0.40%	3.2	Visual
106	Foreign material on threads		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.12	Projectile, Prior to Assembly of Expulsion Charge Assy, Grommet Assy and Prior to Assy of Obturator	1 1	9381131
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special a.	Gap between projectile body and base 360°, max.		100%	3.2	Gage
Major 101	Disassembly torque test of base	(a)	0.40%	3.2	4.5.6
102	Depth from surface of projectile to expulsion charge cup, min.		0.40%	3.2	Gage
103	True position of min. internal diameter of expulsion charge cup to pitch diameter of ogive threads		0.40%	3.2	Gage
104	Expulsion charge cup improperly staked		0.40%	3.2	Visual/Manual
105	Obturator missing or damaged		100%	3.2	Visual
106	Evidence of axial movement or obturator		100%	3.2	Manual
Minor 201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample size per MIL-STD-105, Inspection Level I.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70507 (AR)

PARAGRAPH 4.4.2.13	TITLE Projectile, Prior to Assembly of Grommet Assembly Gasket and Lifting Plug Assembly		SHEET 1 OF 1		DRAWING NUMBER 9381131
					NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special a.	Weight		100%	3.2	Gage
Major 101	Expulsion charge missing		0.40%	3.2	Visual
102	Obturator band broken, loose or missing		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.14	Projectile				9381131
					NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Lifting plug assembly wrong type (not fusible)		100%	3.2	Visual
Special					
a.	Lifting plug or gasket missing or damaged		100%	3.2	Visual
Major					
101	Disassembly torque tests of lifting plug assembly	(a)	0.40%	3.2	4.5.7
102	Air pressure test	(b)	0.40%	3.2	4.5.8
103	Diameter of bourrelet, max (applicable after paint on projectile has been retouched)		100%	3.2	Gage
104	Radiographic examination	(c)		3.5	4.5.9
105	Zone marking missing, incorrect or unidentifiable		0.40%	3.2	Visual
106	Diamond marking missing, incorrect or unidentifiable		0.40%	3.2	Visual
Minor					
201	Any part missing or damaged		0.65%	3.2	Visual
202	Rotating band damaged		0.65%	3.2	Visual

NOTES:
(a) Sample size per MIL-STD-105, Inspection Level I.
(b) Sample size per MIL-STD-1235, CSP-1, Code Letter P.
(c) Ballistic samples only.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.14	Projectile				9381131
					NEXT HIGHER ASSEMBLY
					None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203	Protective coating with bare areas in excess of 1/4 square inch		0.65%	3.2	Visual
204	Marking missing or unidentifiable		0.65%	3.2	Visual
205	Evidence or poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Pallet Assembly	1 OF 1	8837839
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Cover support missing (8 places)		100%	3.2	Visual
Major					
101	Incorrect strapping or improperly assembled		0.40%	3.2	Visual
102	Seal strapping missing		0.40%	3.2	Visual
103	Lifting plug of projectile missing or damaged		0.40%	3.2	Visual
Minor					
201	Board broken or split		0.65%	3.2	Visual
202	NSN number missing or unidentifiable		0.65%	3.2	Visual
203	Any other marking missing incorrect or unidentifiable		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-P-70507 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.16	Obturator		10542907
			NEXT HIGHER ASSEMBLY 9381131

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of rear I.D. with forward I.D.		0.40%	3.2	Gage
102	Diameter outside, front		0.40%	3.2	Gage
103	Diameter outside, rear		0.40%	3.2	Gage
104	Diameter inside, front		0.40%	3.2	Gage
105	Depth of forward inside diameter		0.40%	3.2	Gage
106	Thickness of wall at rear		0.40%	3.2	Gage
107	Length, total		0.40%	3.2	Gage
Minor					
201	Radii or chamfers missing or incorrect		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9362552
4.4.2.17	Lid, Container, Expulsion Charge				NEXT HIGHER ASSEMBLY 9362556
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diamter		0.40%	3.2	Gage
102	Inside diameter of locking lip		0.40%	3.2	Gage
103	Inside diamter of open end		0.40%	3.2	Gage
104	Inside diameter of closed end		0.40%	3.2	Gage
105	Length of cylindrical portion of lid		0.40%	3.2	Gage
106	Incomplete or missing tab		0.40%	3.2	Visual
107	Profile of locking lip area (destructive)	(a)		3.2	Gage
108	Total height of lid		0.40%	3.2	Gage
109	Depth of cap		0.40%	3.2	Gage
110	Thickness of tab below ridges		0.40%	3.2	Gage
111	Width of ribs		0.40%	3.2	Gage
112	Height of ribs		0.40%	3.2	Gage
113	Pull test, min. and max.	(b)		3.2	4.5.12
Minor					
201	Missing slot (4 places)		0.65%	3.2	Visual
202	Presence of friction ribs on tab		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
(a) Sample three (3) parts from each mold cavity at the end of each shift. No failures allowed.
(b) Sample ten (10) per lot: Accept no failures, reject on 1 failure.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Cup, Container, Expulsion Charge	1 OF 1	9362553
			NEXT HIGHER ASSEMBLY 9362555

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of cup		0.40%	3.2	Gage
102	Length of cap interlock area		0.40%	3.2	Gage
103	Outside diameter of ribs		0.40%	3.2	Gage
104	Outside diameter of locking lips		0.40%	3.2	Gage
105	Inside diameter of cup		0.40%	3.2	Gage
106	Outside diameter of cup between ribs		0.40%	3.2	Gage
107	Thickness of bottom		0.40%	3.2	Gage
108	Profile of locking lip area		0.40%	3.2	Gage
109	Pull test, min. and max.	(a)		3.2	4.5.12
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
(a) Sample ten (10) per lot: Accept no failures, reject on 1 failure.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Disc, Expulsion Charge				9362554
					NEXT HIGHER ASSEMBLY 9362555
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Diameter		0.40%	3.2	Gage
Minor					
201	Evidence or poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Container Assembly, Expulsion Charge	1 OF 1	9362555
			NEXT HIGHER ASSEMBLY 9362556

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leak test	(a)		3.2	4.5.2.1
102	Weight of propellant charge incorrect		0.40%	3.2	Gage
103	Disc seal incomplete		0.40%	3.2	Visual
104	Moisture and total volatiles content of propellant	(a)		3.2	4.5.1
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
(a) Sample in accordance with applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9362556
4.4.2.21	Expulsion Charge Assembly				NEXT HIGHER ASSEMBLY 9381131
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cup or lid damaged		0.40%	3.2	Visual
102	Gap between cup and lid 360°, max.		0.40%	3.2	Gage
103	Total length, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.22	Backing plate				9381112
					NEXT HIGHER ASSEMBLY 9381131
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9381113
4.4.2.23	Rubber pad				NEXT HIGHER ASSEMBLY 9381131
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Diameter		0.40%	3.2	Gage
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70507 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9381111
4.4.2.24	Forward plate				NEXT HIGHER ASSEMBLY 9381131
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Largest diameter (2 places)		0.40%	3.2	Gage
102	Diameter of o-ring seat		0.40%	3.2	Gage
103	Width of o-ring seat		0.40%	3.2	Gage
104	Concentricity of o-ring seat		0.40%	3.2	Gage
105	Thickness		0.40%	3.2	Gage
106	Location of basic diameter on taper		0.40%	3.2	Gage
107	Distance from rear face to forward face of o-ring seat		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

DD FORM 68 (H) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing. Testing is required as described herein and in the First Article and Quality Conformance Inspection Table.

4.4.3.1 Ballistic functioning (see 3.4). For first article and for first lot produced and continuing until three (3) consecutive acceptable lots have been accepted, the Government inspector shall select twelve (12) projectiles from each lot for this test. Reduced sampling of six projectiles per lot can commence after three consecutive production lots have been accepted. Rejection of any lot will require resumption of normal sampling until three consecutive production lots are again tested and accepted. They shall be tested per 4.5.10 as follows:

Test Phase	Normal Sampling	Reduced Sampling
Service pressure	8	4
Safety	3	1

4.4.3.1.1 Projectile replacement. The remaining projectile shall be fired for replacement or retest if any of the following occurs:

a. Low bursting rounds. If any service pressure test projectile functions below six hundred (600) feet, and the total number of all grenade defects does not exceed the specified acceptance criteria, the projectile shall be evaluated to function satisfactorily and no replacement for that projectile will be required. However, when the results of the low-bursting projectile would cause the lot represented by the sample to be rejected, the results of that projectile shall be recorded and a replacement projectile shall be fired in the same charge zone. The results of the replacement projectile shall be substituted for those of the low bursting round in the determination of lot acceptance.

b. Lost grenades. A replacement projectile is required whenever more than eight (8) grenades from one projectile during service pressure firing are considered to be lost. Results of these rounds with excessive lost items shall be recorded for informational purposes. The results of the replacement substituted for use in the determination of lot acceptance.

c. Time fuze malfunctions. Time fuze malfunctions shall not be considered as cause for rejection of a projectile lot. When a time fuze malfunction occurs, the proving ground may fire a replacement projectile.

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4.4.3.1.2 Retest. If, for any reason, the proving ground considers that test conditions have affected all or part of the test results, the proving ground shall declare the affected results invalid and perform new tests on additional projectiles as required.

4.4.3.1.3 Rejection. The lot represented by the sample shall be rejected if the ballistic test results for that lot reveal any one of the following defects:

a. Any premature burst occurs (critical defect).
b. Any separation of metal parts occurs prior to initiation of the time fuze (special defect).
c. Any detonation of the projectile cargo occurs prior to initiation of the time fuze (special defect).
d. The total number of grenades failing to function satisfactorily equals or exceeds twenty-nine (29) during normal sampling or fifteen during reduced sampling. (Not applicable to the safety phase) (major defect)
e. Any projectile fails to function satisfactorily (major defect).
f. The total range spread at the point of fuze functioning, when the fuze time is correct exceeds 800 meters during the service pressure test (special defect).

Note 1: A round by round accounting shall be made for lost grenades and the replacement provisions utilized when applicable.

Note 2: Grenade air detonations shall be counted and recorded for informational purposes. However, if the total number of air detonations from the entire eight (8) rounds tested in the service pressure zone phase exceeds thirteen (13) the data shall be referred to the engineering agency (see 6.4) for decision on final acceptance of the lot.

Note 3: Grenades evaluated to have malfunctioned due to damage resulting from air collision/detonation shall not be included in the total number of grenades used to accept or reject the lot. Grenades so malfunctioning shall be counted and recorded on the evaluation sheet for informational purposes. However, if the total number of such grenades from the entire sample tested in the service pressure phase exceed sixteen (16) during normal sampling or eight (8) during reduced sampling, the data shall be referred to the engineering agency (see 6.4) for decision on final acceptance of the lot.

4.4.3.3 Dimensional control of plastic parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (see 4.4.3.3.1), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

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The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection or control of each cavity during production shall be performed on the above quantities from each cavity after production of each 500 aparts or at the end of each shift, providing the number of parts produced during the shift is less than 500 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective parts will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault may, after reworking, be returned to production providing they pass the qualifiction test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

The contractor shall furnish evidence to the Government inspector that dimensional stability has been attained prior to measuring the parts for acceptance.

4.4.3.3.1 Curing time. To establish a curing time dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time that part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Commander, US Army AMCCOM, ATTN: AMSMC-QAR-Q(D), Dover, NJ 07801-5001. If any new change in material, cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.4.4 Inspection equipment. For the inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078, paragraph 6.3.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

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4.4.4.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410.

4.5 Test methods and procedures.

4.5.1 Moisture and total volatiles content of propellant. The moisture and total volatiles content of each lot of propellant or portion thereof shall be determined for each eight (8) hours of production. This determination shall be accomplished prior to the anticipated use of the lot or portion thereof, with a maximum of eighteen (18) hours lead time. Sampling of the propellant shall be in accordance with MIL-STD-286, Method 101.1. If the moisture and total volatiles content fails to comply with the requirements of MIL-P-48099, the propellant lot shall be rejected until corrective action has been taken as verified by repeating the moisture and total volatiles content determination. The contractor shall provide adequate controls to insure maintaining compliance with the requirements between testing and use.

4.5.2 Leak test of expulsion charge assembly (destructive test). A sample of five (5) assemblies from each hour of production per sealing machine shall be selected for this test. Each sample assembly shall be subjected to the vacuum specified on the applicable drawing using equipment and procedures shown on drawings 884732 and 9202990 (except test instruction 4, see Note 1 for reject criteria). If any sample assembly fails the leak test, (see note 1) the quantity represented by that sample shall be set aside and retested per the following procedure:

a. Visually inspect all bags for completeness of heat seals (top, side, and bottom). Bags with damaged or incomplete seals shall be classed defective and removed from production.

b. From remaining assemblies, ten (10) shall be selected and subjected to the leak test, accept on 0 defects, reject on 1 defect.

Note 1: For the purpose of this test a reject shall be defined as an assembly that leaks to the extent that the M10 propellant becomes wet. To determine the condition of the propellant those assemblies that exhibit bubbles during the test shall be cut open and the propellant examined (visually and manually) for moisture. Only those assemblies containing moisture shall be considered defective.

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4.5.2.1 Leak test of expulsion charge container assembly (dwg. 9362555) (destructive test). A sample of five (5) assemblies (without lids) from each hour's production per sealing machine shall be selected for this test. Each sample assembly shall be subjected to the vacuum specified on the applicable drawing using equipment and procedures shown on drawings 8841732 and 9202990 (except test instruction 4, see Note 1 for reject criteria). If any sample assembly fails the leak test (see note 1), the quality represented by that sample shall be set aside and retested per the following procedure:

a. Visually inspect all cups for completeness of heat seals. Cups with damaged or incomplete seals shall be classed defective and removed from production.

b. From remaining assemblies, ten (10) shall be selected and subjected to the leak test; accept on 0 defect, reject on 1 defect.

Note 1: For the purposes of this test, a reject shall be defined as an assembly that leaks to the extent that the M10 propellant becomes wet. To determine the condition of the propellant those assemblies that exhibit bubbles during the test shall be cut open and the propellant examined (visually and manually) for moisture. Only those assemblies containing moisture shall be considered defective.

4.5.3 Strength of heat seal. This test shall be performed in accordance with FED-STD-101, Method 2024, as modified herein. One specime, one inch wide, shall be cut perpendicular to the line of a seam of each sample bag. Location of the sample shall be random so that both the end seam and the long seam are sampled. One leg of the specimen shall be clamped so that the specimen hangs freely. The required load (2.5 lbs) shall be applied to the free end in such a manner as to avoid impact loading the seam. The load shall be allowed to act for a minimum of one minute after which the seam shall be examined for peel-back of the heat seal.

Note: The test samples shall be conditioned prior to testing at a temperature of 70 to 80 degrees Fahrenheit (F) and shall be tested under these conditions unless otherwise specified.

4.5.4 Verification of heat treatment of adapter (non-destructive). This inspection shall employ the eddy current method, or any other equivalent method approved by the procurement agency. If the eddy current method is used, all personnel operating and calibrating this equipment shall be qualified and certified in accordance with MIL-STD-410. The hardness limits shall be established and submitted to the COR for approval prior to use. If any adapter fails to comply with the requirements specified on the applicable drawing, it shall be classed defective and removed from production.

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4.5.5 Hardness test of shim (dwg. 9215397) (non-destructive). This test shall be performed in accordance with ASTM E 18 using the standard penetrator for the specified hardness scale. Three readings shall be taken approximately 120° apart. If any reading fails the requirement on the applicable drawing that shim shall be classed defective.

4.5.6 Disassembly torque test of base. The projectile with base assembled, shall be subjected to the disassembly torque specified on the applicable drawing. Any unit that fails to withstand the disassembly torque shall be classed defective.

4.5.7 Disassembly torque test of lifting plug assembly. The projectile with lifting plug assembled, shall be subjected to the disassembly torque specified on the applicable drawing. Any unit that fails to withstand the disassembly torque shall be classed defective.

4.5.8 Air pressure test of projectile. The assembled projectile, with lifting plug, shall be placed in one or more chambers, (the base to body, base to base closure, base closure seal, body to ogive and ogive to lifting plug joints must be tested) and a pre-determined volume of air shall be delivered to produce the specified pressure between the inside of the chamber and the outside of the projectile. After initiation of test, the pressure must achieve the required level and maintain that level for the specified time interval. Failure to meet either criteria shall be cause for rejection and that projectile shall be suitably identified and removed from the lot to prevent mixing with accepted projectiles.

All equipment used in this test shall be capable of performing the test accurately and consistently. Equipment shall be calibrated at the start of each shift and thereafter, at intervals of not more than four (4) hours of continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of three (3) hours or more. If it is found that the gage is out of calibration in such a direction that the test pressure was actually lower than that specified on the drawing, projectiels tested since the last satisfactory calibration shall be retested after the gage error has been corrected.

4.5.9 Radiographic inspection of projectile. The projectile assemblies shall be radiographically examined prior to forwarding to the proving ground. Prior to assembly of the base burner assembly, the Government inspector shall indicate the location of the spline by marking the outside of the projectile. In all exposures the radiographic beam shall be focused along the center plate, with the spline ninety degrees (90°) to the right in the direction from the source to the projectile. Radiographic exposure shall as a minimum, include all the base material

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adjacent to the adapters, to and including the entire forward plate. If necessary, more than one radiograph shall be taken in this view as required to avoid detrimental parallax effects and to observe the conditions specified in 3.5. In the event the radiograph(s) taken in this view prevent a suitable examination, the projectile shall again be radiographed at a location rotated 90° from the original with the spline nearest the source. Projectiles and radiographs shall be suitably identified to permit cross reference. Radiographs shall be maintained by the contractor until final acceptance of the projectile lot.

If any defect, per paragraph 3.5 is found, it shall be reported to AMSMC-QAR-Q(D). The lot represented by this sample shall be rejected.

4.5.9.1 Radiographic qualifications. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. The personnel performing this examination shall be qualified and certified, level 2, per MIL-STD-410. In addition, the radiographic examination shall be performed with technique capable of producing 2 percent sensitivity (radiographic quality level 2-2T) and the photographic density of the film used shall be of a quality which will produce a sharp definition of internal structure.

4.5.10 Ballistic functioning tests. These test shall be conducted at a Government proving ground in accordance with an applicable acceptance test procedure. The round shall be fitted with a M577 fuze. The projectile fuze setting and quadrant elevation (QE) of the weapon shall be adjusted to attempt to achieve ejection, at the target height specified below, over a prepared impact area. During any test phase the fuze setting and QE will remain constant. The maximum ejection height shall be eighteen hundred (1800) feet, for any zone. A photograph shall be taken of the fuze setting for each round immediately prior to use and identified with the round number. The sample projectile shall be tested at the specified zones using weapons, equipment methods and procedures in accordance with the acceptance test procedures referenced above. Tubes used for this test shall not have less than twenty-five (25) percent remaining tube life based on current gun tube condemnation criteria.

4.5.10.1 Service pressure. The sample projectiles to be subjected to this test shall be fired in the M199 Cannon, using the M203 series propelling charge. The target ejection height for this phase shall be fifteen hundred (1500) feet. Observation shall be made for failure to comply with the requirements of 3.4.

4.5.10.2 Safety phase. The sample projectiles to be subjected to this test shall be fired at ambient temperature in the M199 Cannon after modification of the M203 propelling charge by adding sufficient propellant to yield a chamber pressure of

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60 to 64 thousand pounds per square inch (KPSI). Observation shall be made for failure to comply with the requirements of 3.4.3. Burst height and grenade functioning are not considered requirements of this test, but shall be reported for informational purposes.

4.5.10.4 Records of ballistic tests. A record shall be made for each round fired and shall include an accounting for the defects listed. In addition, a record shall be made of fuze duds, fuzes with excessive off-times, and any other malfunctions. A copy of the records of all ballistic test results shall be forwarded to the address specified in 6.5.

4.5.11 Depth to adapters or shims. This measurement shall be performed using automatic or semi-automatic inspection equipment. The design of this equipment shall be "fail-safe" to the extent that the projectile cannot be released for further processing until the proper number of shims have been inserted. Verification that the proper number of shims have been inserted shall be accomplished by re-inspection of this depth.

4.5.12 Pull test of empty plastic expulsion charge assembly (destructive test). After each sample of the lid (dwg. 9362552) and the empty cup (dwg. 9362553) have been assembled together for this inspection, the assembly shall then be subjected to this test by placing it in a suitable fixture which will restrain the cup in such a manner as to avoid any squeezing on the body of the cup. The load, in an axial direction, shall then be applied to the tab of lid, and the force required to separate the lid from the cup shall be measured. If any assembly fails to meet the minimum and maximum force requirements on the applicable drawing (9362556) the production quantity, both cup and lid, represented by that sample shall be rejected. All equipment used to perform this test shall be in accordance with 4.4.4.

5. PACKAGING

5.1 Packing

5.1.1 Level A. The projectiles shall be palleted in accordance with Dwg. 8837839.

5.2 Marking. Marking shall be in accordance with Dwg. 8837839 and Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

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6.1 Intended use. Not applicble to this specification.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, USA AMCCOM ATTN: AMSMC-QAR-I(D), Dover, NJ 07801-5001.

6.4 Submission of ballistic test data and data cards. In addition to the normal distribution of records, when the projectile is procured by the U.S. Army Armament Material and Readiness Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, USA AMCCOM, ATTN: AMSMC-QAR-Q(D) Dover, NJ 07801-5001.

6.5 Data requirements. Data item description required for submission under this specification are as follows:

Paragraph	Data requirement	Applicable D.I.
4.5.9, 4.5.10	Quality Inspection Defect Report	DI-R-1721

The above data shall be forwarded in accordance with the requirements of the contract to Commander, USA AMCCOM, ATTN: AMSMC-QAR-Q(D), Dover, NJ 07801-5001.

6.6 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

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Preparing Activity:
Army - AR

(Project 1320-A819)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70507	PROJECTILE, 155MM, HE, XM864 LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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